Springboard Virtual Internship 6.0

Name: Akshata.T.Rathod

College: Jawaharlal Nehru New College of Engineering

Branch: Computer Science and Engineering

Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

Mentor: G.K.S Jyoteesh

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Milestone-03: Geographical Analysis of Disaster Declarations

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1. Objective

The objective of this milestone is to analyze and visualize the **geographical distribution of disaster declarations across the United States** using the FEMA disaster declarations dataset.

This milestone focuses on identifying **geographical patterns and disaster hotspots** by analyzing disaster frequency across states and regions. Spatial visualizations such as **choropleth maps, state comparison charts, and stacked bar charts** are used to better understand how disasters vary geographically.

Through this analysis, we aim to identify regions that are more vulnerable to disasters and understand how different types of disasters are distributed across states.

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2. Problem Formulation

Analysis Type

This milestone involves the following types of analysis:

- Geospatial Data Analysis
- Exploratory Data Analysis (EDA)

The goal is to transform disaster declaration data into spatial visualizations to uncover geographic trends.

Key Questions

The following questions guide the geographic analysis:

- Which states experience the highest number of disaster declarations?
- Are disasters concentrated in specific regions of the United States?
- Do certain disaster types occur more frequently in specific states?
- Can geographic hotspots of disaster activity be identified?

These questions help in understanding the **spatial distribution of disasters**.

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3. Dataset Preparation for Geographic Analysis

Input Dataset

The dataset used for this analysis is:

usnd_cleaned.csv

This dataset is the cleaned dataset generated during **Milestone-01 data preprocessing**.

Required Columns Used

The following columns were used for the geographic analysis:

- **state** – Represents the state where the disaster occurred
- **incidentType** – Type of disaster (e.g., Flood, Hurricane, Fire)
- **year** – Year of the disaster declaration
- **month** – Month of the disaster declaration

Additional optional columns include:

- **declarationDate**
- **stateName**

Data Validation

Before performing geographic analysis, the dataset was validated to ensure data consistency.

The following checks were performed:

- Verified that state codes were consistent
- Checked for missing values in the state column
- Ensured that disaster records exist across multiple states

These steps ensured that the dataset was suitable for geographic aggregation and spatial analysis.

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4. Geographic Aggregation

Geographic aggregation was performed to analyze disaster declarations by location.

4.1 State-Level Aggregation

The dataset was grouped by **state** to calculate the total number of disaster declarations in each state.

This aggregation helps identify:

- States with the highest disaster frequency
- Geographic distribution of disaster events

Example aggregated result:

State	Disaster Count
Texas	520
California	480
Florida	460

This aggregation forms the basis for geographic visualizations such as the choropleth map.

4.2 State + Incident Type Aggregation

The dataset was also grouped by:

- State
- Incident Type

This aggregation helps determine **which disaster types dominate in specific states**.

Example result:

State	Incident Type	Count
California	Fire	150
Florida	Hurricane	180

This information is later used in the **stacked bar chart visualization**.

4.3 Regional Comparison (Optional)

States were grouped into four major regions of the United States:

- West
- Midwest

• South
• Northeast

Regional aggregation helps identify broader geographic patterns in disaster frequency.

This analysis provides insight into how disasters vary across larger geographic areas.

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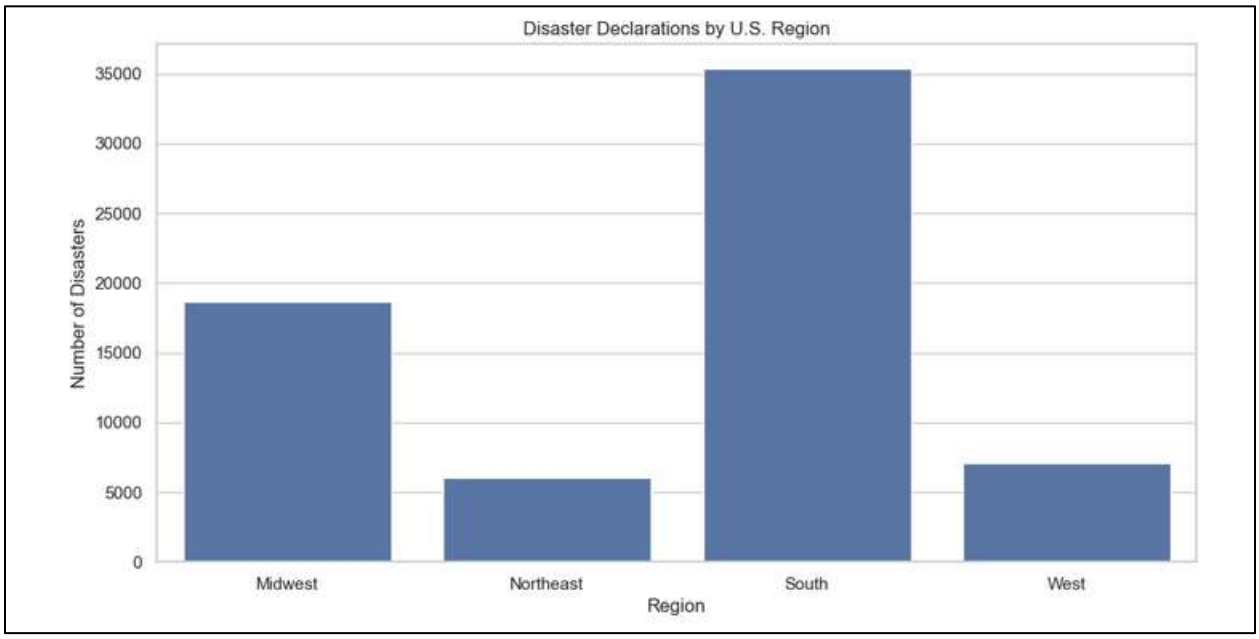
5. Visualizations

Several geographic visualizations were created to analyze disaster distribution.

5.1 Regional Comparison Chart



Interpretation

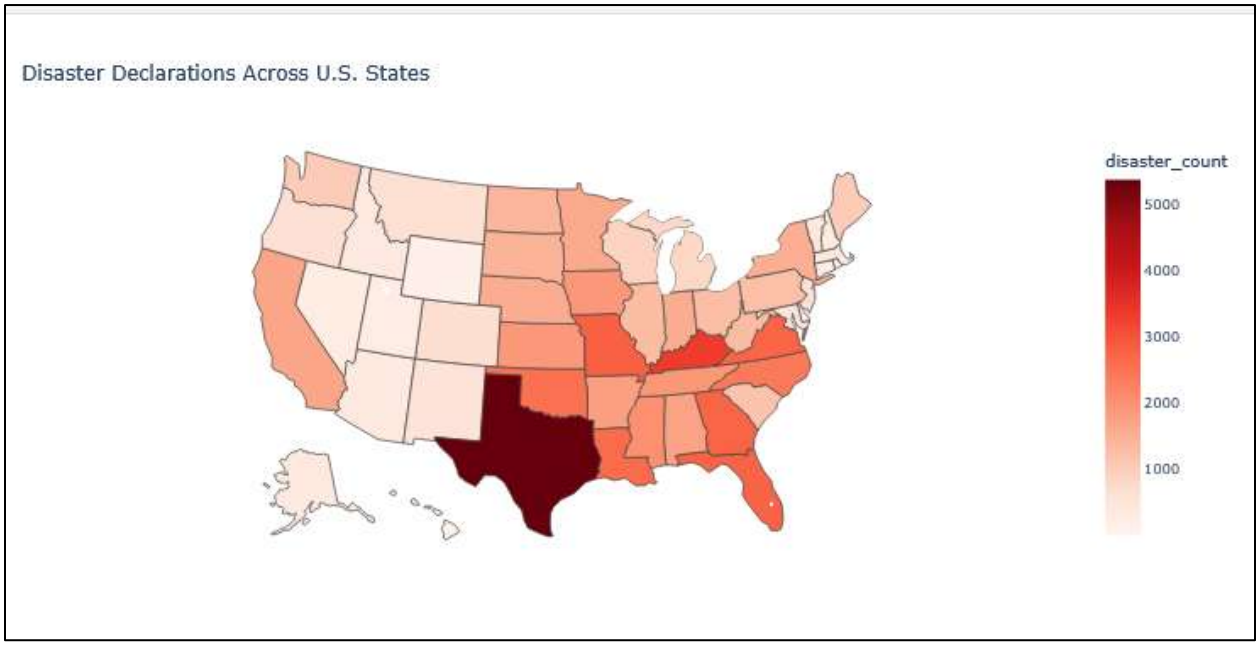
This visualization compares disaster declarations across the four major regions of the United States.

The **Southern region** shows the highest disaster frequency due to exposure to hurricanes, floods, and tropical storms.

Western states also experience a significant number of disasters, particularly wildfires.

The Midwest and Northeast regions show comparatively lower disaster frequencies.

5.2 Choropleth Map – Disaster Distribution by State



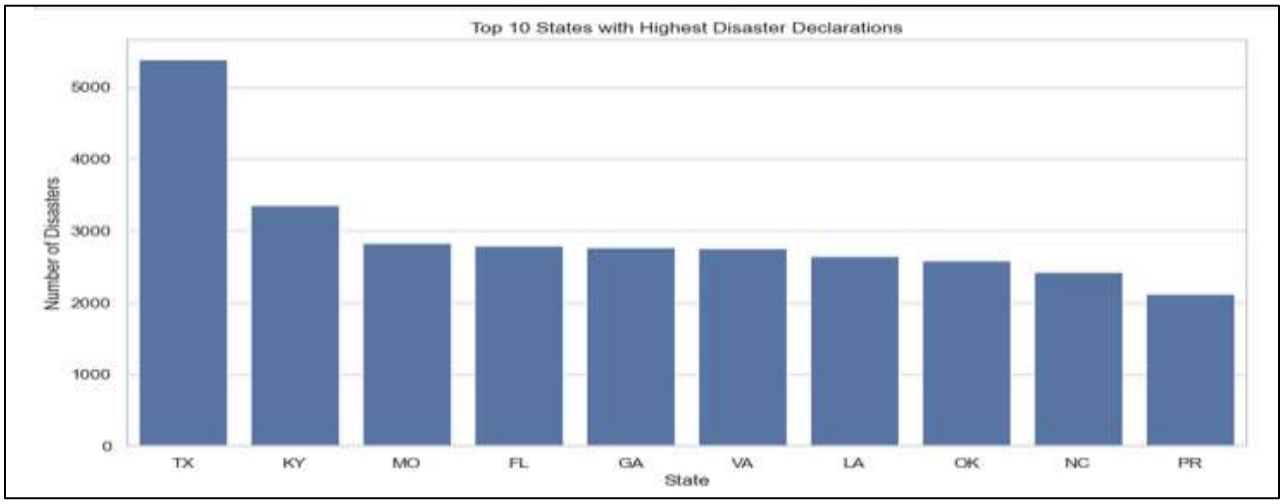
Interpretation

The choropleth map visualizes the spatial distribution of disaster declarations across U.S. states.

States with darker colors represent higher disaster frequencies. Southern and coastal states such as **Texas and Florida** appear to have higher disaster counts.

This pattern reflects the impact of hurricanes, storms, and flooding events in these regions.

5.3 State Comparison Chart (Top 10 States)



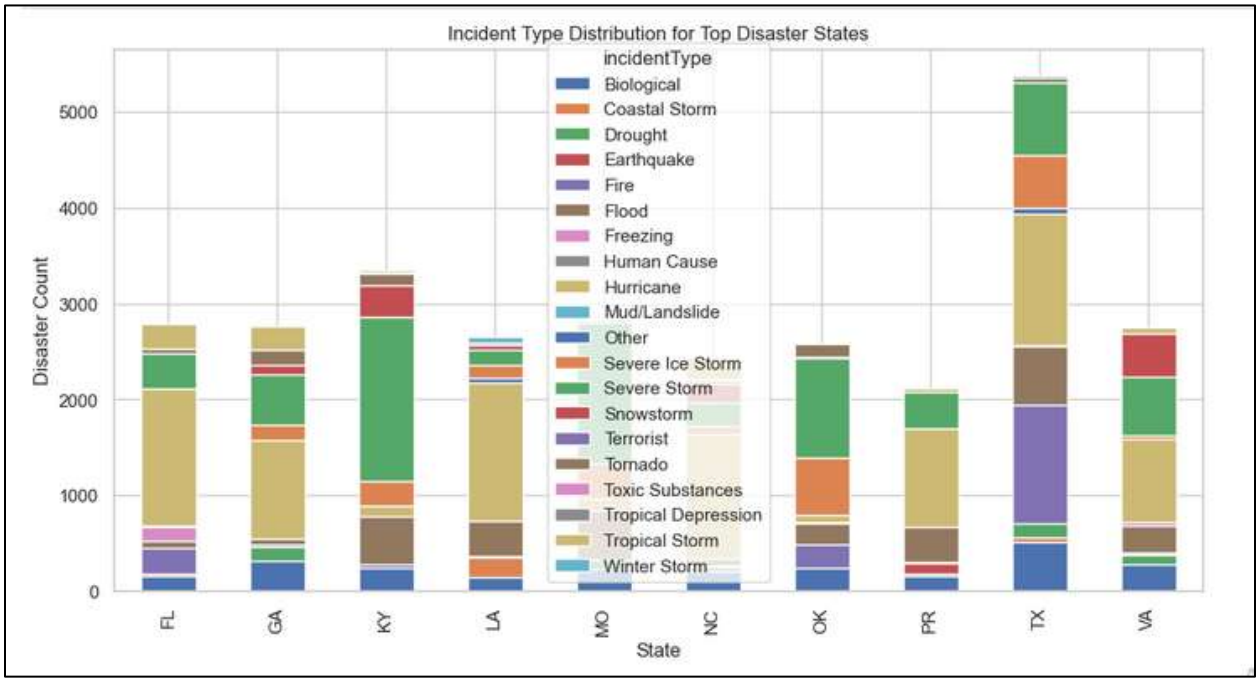
Interpretation

This bar chart compares the states with the highest number of disaster declarations.

A small number of states dominate the dataset, indicating significant geographic differences in disaster exposure.

States such as **Texas, California, and Florida** frequently appear among the most disaster-prone states.

5.4 Incident Type Distribution by State



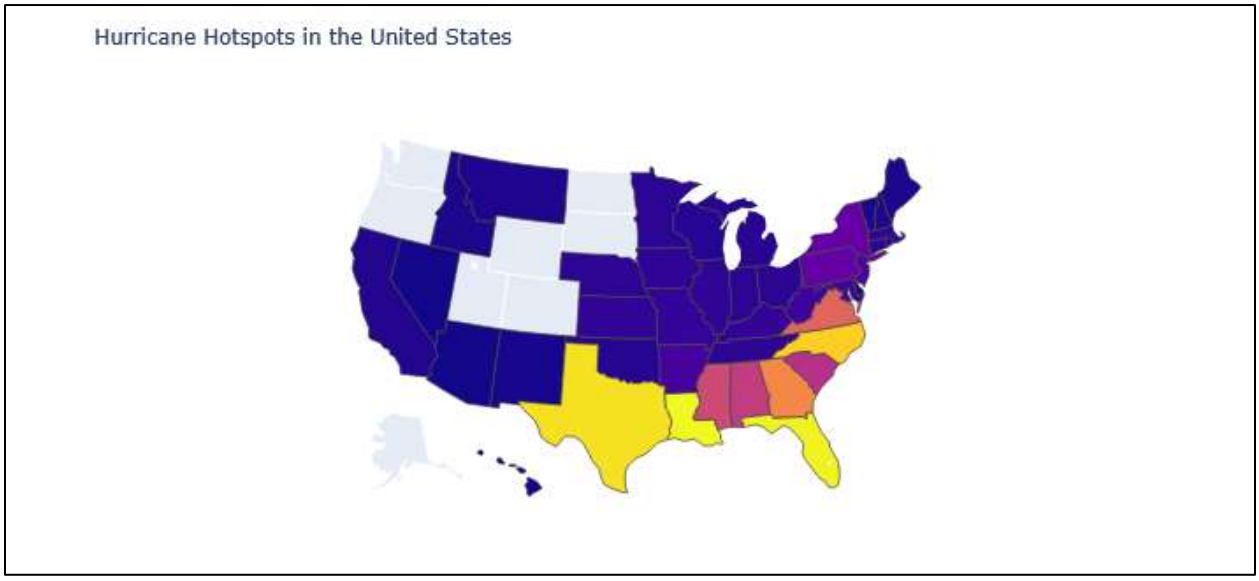
Interpretation

This stacked bar chart illustrates how different disaster types contribute to total declarations within each state.

Certain disaster types dominate specific states depending on their geographical and climatic conditions.

For example, wildfires are common in western states, while hurricanes are more frequent in coastal regions.

5.5 Hurricane Hotspot Map



Interpretation

This map highlights hurricane-related disaster declarations across states.

Coastal states along the **Gulf of Mexico and Atlantic Ocean** show higher hurricane disaster frequencies.

This visualization helps identify regions that are particularly vulnerable to hurricanes.

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6. Tools and Libraries Used

The following tools and libraries were used for the analysis.

Data Processing

- **Pandas** – Used for data loading, cleaning, and aggregation
- **NumPy** – Used for numerical operations and data handling

Visualization

- **Matplotlib** – Used to generate basic plots and charts
- **Seaborn** – Used for visually enhanced statistical charts
- **Plotly** – Used for creating interactive geographic maps

These libraries enabled both statistical and spatial analysis of disaster data.

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7. Key Insights

The geographic analysis reveals several important patterns:

- Disaster declarations are not evenly distributed across the United States
- Southern and coastal states experience higher disaster frequencies
- Certain disaster types are concentrated in specific regions
- Geographic visualizations clearly highlight disaster hotspots

These insights demonstrate the importance of geographic analysis in disaster risk assessment.

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8. Conclusion

Milestone-03 successfully analyzed the **geographical distribution of disaster declarations** across the United States.

Through geographic aggregation and spatial visualizations, the analysis identified disaster-prone states, regional patterns, and disaster hotspots.

The results show that disaster occurrences are strongly influenced by geographic and climatic factors.

This milestone builds upon previous analysis and provides deeper insights into the spatial characteristics of natural disasters.